sysMonitor AMQP/TLS/JSON

Interface Design Description

Service ID: *"sysMonitor"*

Abstract
This document describes the AMQP 0-9-1 protocol with TLS payload security and JSON payload encoding variant of the `sysMonitor` service of the LocalCloudManager system (SysMonitor-AMQP-IDD-5.2.2).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the AMQP binding of the native `sysMonitor` service of LocalCloudManager.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	AMQP	0-9-1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `sysMonitor` service interface

This document provides the Interface Design Description IDD to the *sysMonitor – Service Description* document [5].

2.1 Interface registration

With `sysmonitor.enabled=true`, the common sysMonitor publisher of LocalCloudManager registers the service at start-up with interface template `generic_amqps` (protocol `amqps`, policy `CERT_AUTH`; `generic_amqp` without TLS) and the following properties and metadata:

```
{
  "serviceDefinitionName": "sysMonitor",
  "version": "1.0.0",
  "metadata": {
    "contract": "SysMonitor-AMQP-IDD-5.2.2",
    "schemaVersion": "1.0",
    "publisherMode": "native",
    "observationTypes": [
      "echo",
      "power",
      "status"
    ],
    "delivery": "persistent, publisher-confirmed"
  },
  "interfaces": [
    {
      "templateName": "generic_amqps",
      "protocol": "amqps",
      "policy": "CERT_AUTH",
      "properties": {
        "accessAddresses": [
```

```
          "amqp-broker"
        ],
        "accessPort": 5671,
        "exchange": "arrowhead",
        "baseRoutingKey": "TestCloud.sysmonitor.LocalCloudManager.LocalCloudManager",
        "operations": [
          "echo",
          "power",
          "status"
        ]
      }
    }
  ]
}
```

Listing 1: Registered service instance (TestCloud).

3 Service Operations

The publisher is the `SysMonitorPublisher` of the vendored Eclipse Arrowhead common library (RabbitMQ Java client), configured with the `sysmonitor.*` Spring properties (`enabled`, `broker.address`, `broker.port`, `local.cloud`, `service.instance`, `publish.interval`, `ssl.*`). The echo health comes from the Docker health indicator: degraded while the Docker proxy cannot be reached; `serviceCount` is the number of services in the system information (2). The publisher connects to the broker (default `amqp-broker`, port 5671) with TLS, verifies the broker host name and authenticates with its X.509 system certificate (SASL `EXTERNAL`; the certificate CN is the broker user name). Every publication interval it publishes each observation to the topic exchange `arrowhead` with the routing key `<baseRoutingKey>.<operation>`, where the base routing key is `<Cloud>.sysmonitor.<System>.<serviceInstance>`. Messages are persistent (delivery mode 2), `application/json`, carry the observation identifier as message id, and are publisher-confirmed (timeout 5 s); a failed cycle is retried with a new connection at the next interval. The broker's topic permissions allow the system to publish only below its own base routing key.

3.1 PUBLISH `<Cloud>.sysmonitor.LocalCloudManager.LocalCloudManager.echo`

Operation: echo
Output: Observation

Liveness: `status` (`alive` or `degraded`), `uptimeSeconds` and a `detail` text; quality `good` or `uncertain`.

3.2 PUBLISH `<Cloud>.sysmonitor.LocalCloudManager.LocalCloudManager.power`

Operation: power
Output: Observation

Power supply: `supplyType=line`, `state=available`; unit `1`.

3.3 PUBLISH `<Cloud>.sysmonitor.LocalCloudManager.LocalCloudManager.status`

Operation: status
Output: Observation

Process state: `state=running`, `health` (`healthy`/`degraded`) and `serviceCount`, the number of registered services.

```
{
```

```
  "schemaVersion": "1.0",
  "observationId": "LocalCloudManager-echo-1792670400000-42",
  "providerSystem": "LocalCloudManager",
  "serviceInstance": "LocalCloudManager",
  "observationType": "echo",
  "sourceTime": "2026-10-22T12:00:00.000Z",
  "sequence": 42,
  "value": {
    "status": "alive",
    "uptimeSeconds": 3600,
    "detail": "LocalCloudManager is running"
  },
  "quality": "good",
  "metadata": {
    "publisherMode": "native"
  }
}
```

Listing 2: An echo observation.

4 Data Models

4.1 struct Observation

Object Field	Value Type	Description
"schemaVersion"	String	`1.0`.
"observationId"	String	`<System>-<type>-<epoch ms>-<sequence>`.
"providerSystem"	String	The publishing system.
"serviceInstance"	String	Service instance, default the system name.
"observationType"	String	`echo`, `power`, `status`.
"sourceTime"	String	ISO 8601 UTC time with milliseconds.
"sequence"	Number	Sequence number per publisher.
"value"	Object	Observed value, see the operations.
"unit"	String	`1` for `power` and `status`; absent for `echo`.
"quality"	String	`good` or `uncertain`.
"metadata"	Object	`{"publisherMode": "native"}`.

4.2 Primitives

Type	Description
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

5 Go and Rust implementation variants

5.1 Go implementation

Primary system folder: `Eclipse-Arrowhead-go/ah5-LocalCloudManager-go`.

The system owns its Go module, launcher and build configuration. Use the per-system `Makefile` and `go.mod`; the containing Go tree is the build source, not a historical Arrowhead copy. The native manager has its own browser/session, provider-adapter and optional telemetry profile. Its detailed contracts are in `Eclipse-Arrowhead-go/ah5-LocalCloudManager-go/documentation/v5.3.0`. Secure and legacy profiles must be distinguished; a console cookie is not an Arrowhead identity token.

Listening port. Default 8480 on all interfaces. Override with SERVER_PORT=N (or PORT=N) or –server.port N. Native design/route evidence: `Eclipse-Arrowhead-go/ah5-LocalCloudManager-go/documentation/LocalCloudManager`.

For this Java interface, select only a native transport and operation explicitly implemented and advertised by that port. A matching method/path does not establish DTO, error, authorization, retry or persistence equivalence. Java MQTT/AMQP bindings are not automatically provided by a Go or Rust role. Differential contract and security checks remain necessary before substitution.

5.2 Rust implementation

Primary system folder: `Eclipse-Arrowhead-rust/ah5-LocalCloudManager-rust`.

The manager has its own Cargo crate and legacy adapter profile; it is not the shared support runtime's secure binding. Its listener/provider authentication and unsupported mutations are documented in `Eclipse-Arrowhead-rust/ah5-LocalCloudManager-rust/documentation/v5.3.0`. Build from the primary Rust workspace with the applicable package/binary selection. Documentation links can resolve into the shared core/support catalogue. Independent process identity and state remain required.

Listening port. System API default 127.0.0.1:8480. Override with "listen" in CONFIG.json, SERVER_PORT=N or –server.port N. The HTTPS console listener ("consoleListen") is configured separately. Native design/route evidence: `Eclipse-Arrowhead-rust/ah5-support-runtime-rust/documentation/LocalCloudManager`.

For this Java interface, select only a native transport and operation explicitly implemented and advertised by that port. A matching method/path does not establish DTO, error, authorization, retry or persistence equivalence. Java MQTT/AMQP bindings are not automatically provided by a Go or Rust role. Differential contract and security checks remain necessary before substitution.

6 References

[1] AMQP Working Group, “Amqp 0-9-1 specification,” AMQP Working Group, Tech. Rep., 2008.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “sysMonitor – Service Description (LocalCloudManager),” Eclipse Arrowhead, SD 5.3.0, 2026, localCloudManager-v5.3.0-sysMonitor-SD.pdf.

7 Revision History

7.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)
3	2026-11-09	5.2.2	Java-Spring implementation: the observations are published by the common-library `SysMonitorPublisher` (RabbitMQ Java client) with the `sysmonitor.*` properties; health from the Docker proxy; wire format unchanged.	Claude Opus 5.5 (draft)
4	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

7.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	